DYNAMICAL SYSTEMS:
CLASS SEPTEMBER 8, 2026
5A. 2D LINEAR SYSTEMS: PHASE PLANE

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Teams

Post screenshots of your work to the course Google Drive today: Link to drive folder. Include words, labels, and other short notes on your whiteboard that might make those solutions useful to you or your classmates. Name your images using the format:

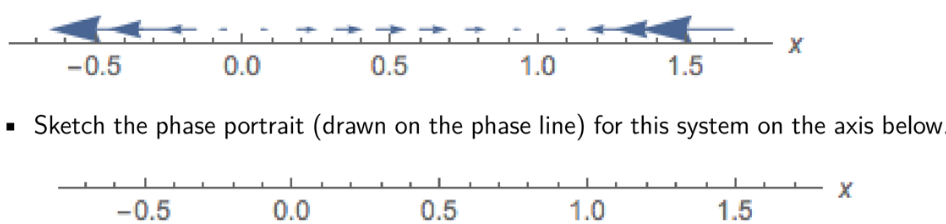
- `teamXX-YY.EXT`

where XX is your two-digit team number (e.g. 06 if you are team 6), YY is the two digit number of image for today (e.g. 01 for the first image, 02 for the second, and so on.), and EXT is the extension for the image filetype you used.

Assigning roles

- The person who has most recently traveled outside of Richmond will be the time-keeper (keeping the team on task and making sure that team members are taking turns contributed / sharing time).
- The next-most-recent person will be scribe the first problem (doing all writing for the team).
- Choose a remaining team member to be the questioner (checking in to make sure that team members are asking any questions that they have).

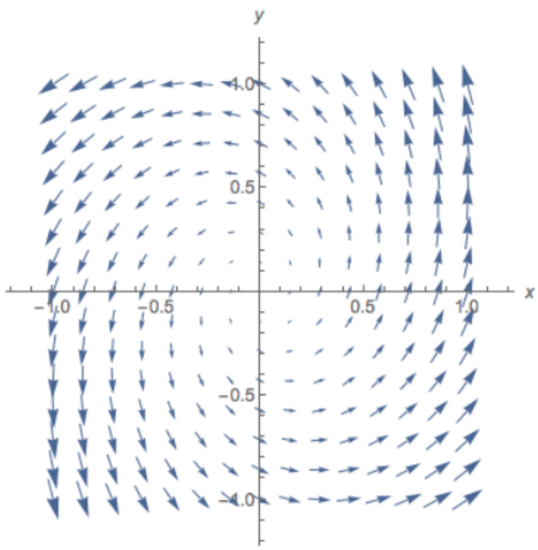
(1) (1d vs 2d)
  (a) Consider the dynamical system $\dot{x} = f(x)$ with $f(x) = x - x^2$.
  Here is a plot of the vector field. The vector field is an assignment of the vector $f(x)\vec{i}$ to the point x.

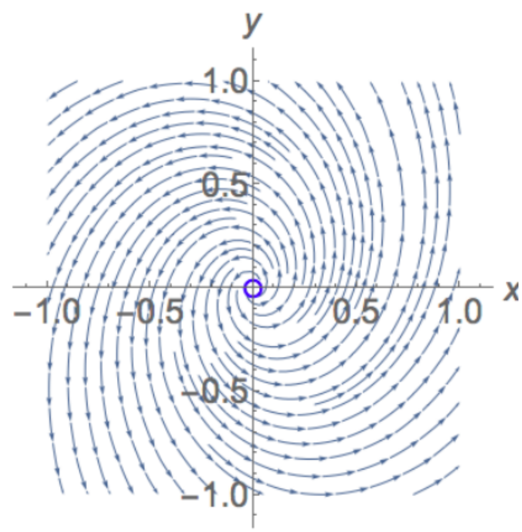


  - Sketch the phase portrait (drawn on the phase line) for this system on the axis below.
  - Identify how the phase portrait is similar to or different from the vector field.
  - How do trajectories appear in the 1d phase portrait?

Date: September 8, 2026.

(b) Now consider the dynamical system $\begin{array}{l} \dot{x} = f(x,y) \\ \dot{y} = g(x,y) \end{array}$. The vector field is an assignment of the vector $f(x,y)\vec{i} + g(x,y)\vec{j}$ to the point (x,y). Let $f(x,y) = x - 2y$, $g(x,y) = 3x + y$. Consider the two images below. Which one is the vector field (plotted in the phase plane), and which one is the phase portrait (drawn on the phase plane)?





Identify how the phase portrait is similar to or different from the vector field.

Solution. (a) We factor the right-hand side:

$$\dot{x} = x - x^2 = x(1-x).$$

Thus the equilibria are $x = 0$ and $x = 1$. The sign of $\dot{x}$ on the three complementary intervals is

$$\begin{array}{c|ccc} x & (-\infty,0) & (0,1) & (1,\infty) \\ \hline x(1-x) & - & + & - \end{array}.$$

Therefore the phase line is

$$(-\infty) \longleftarrow \boxed{0} \longrightarrow \boxed{1} \longleftarrow (+\infty).$$

The equilibrium $x = 0$ is unstable because arrows point away from it, while $x = 1$ is asymptotically stable because arrows point toward it.

The vector field and the phase portrait contain the same directional information: the sign of $f(x)$ tells whether motion is to the left or right, and the zeros of f give the equilibria. The vector-field plot displays a vector at each sampled point, often with its length reflecting $|f(x)|$. The phase portrait emphasizes the solution orbits and their direction, so arrow lengths are not essential.

In one dimension, trajectories lie on the phase line. A solution beginning in $0 < x < 1$ moves monotonically right toward 1, and a solution beginning with $x > 1$ moves monotonically left toward 1. A solution beginning with $x < 0$ moves left, away from 0. The two equilibrium trajectories remain fixed at $x = 0$ and $x = 1$.

Non-equilibrium trajectories cannot cross either one. For initial value $x(0) = x_0$, the non-equilibrium solutions can also be written

$$x(t) = \frac{x_0 e^t}{1 + x_0(e^t - 1)}$$

on their maximal intervals of existence.

(b) In the displayed pair, the **left-hand image is the vector field**: it consists of many individual arrows showing the velocity assigned at sampled points. The **right-hand image is the phase portrait**: it shows solution curves whose tangent vectors agree with the vector field.

To verify the qualitative shape, write the system as

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & -2 \\ 3 & 1 \end{pmatrix}.$$

Its characteristic polynomial and eigenvalues are

$$\det(A - \lambda I) = (1 - \lambda)^2 + 6, \qquad \lambda = 1 \pm i\sqrt{6}.$$

Because the real part is positive, the origin is an unstable spiral: nearby nonzero trajectories spiral outward as time increases. At $(1, 0)$ the velocity is $(1, 3)$, which points upward, so the rotation is counterclockwise. This agrees with the right-hand phase portrait.

$\square$

(2) (Generic 2d system of linear differential equations)

Rotate roles: questioner → timekeeper → scribe

Consider the system

$$\frac{dx}{dt} = ax + by$$
$$\frac{dy}{dt} = cx + dy$$

with fixed point at the origin.

(a) Rewrite this in matrix / vector form (let $\underline{x} = \binom{x}{y}$.)

(b) Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Recall that the eigenvalues of A, λ_1 and λ_2, are given by $\lambda^2 - \tau\lambda + \Delta = 0$ where $\tau = a + d$ is the trace of the matrix A and $\Delta = ad - bc$ is the determinant of the matrix A. In addition, $(\lambda - \lambda_1)(\lambda - \lambda_2) = 0$. Why?

(c) Use this second fact (and match terms in the two equations) to show that $\lambda_1 + \lambda_2 = \tau$ and $\lambda_1\lambda_2 = \Delta$.

(d) Consider the $\Delta\tau$-plane (with Δ on the horizontal axis). Identify regions of the $\Delta\tau$ plane where the matrix has two negative eigenvalues, regions where it has one positive eigenvalue and one negative one, and regions where it has two positive eigenvalues.

(e) Consider $\underline{v}e^{\lambda_1 t}$, where λ_1 is an eigenvalue of A and $\underline{v}$ is the associated eigenvector. Show that this is a solution of the differential equation (Hint. refer back to the example in Lecture 5 on linear systems). Think about plotting this solution as a trajectory in the xy-plane. For λ_1 a real number, why is it a straight-line solution?

(f) General solutions to this differential equation are of the form $\underline{x}(t) = c_1\underline{v}_1 e^{\lambda_1 t} + c_2\underline{v}_2 e^{\lambda_2 t}$. We might have λ_1 and λ_2 both real. Or they could be a complex conjugate pair, where

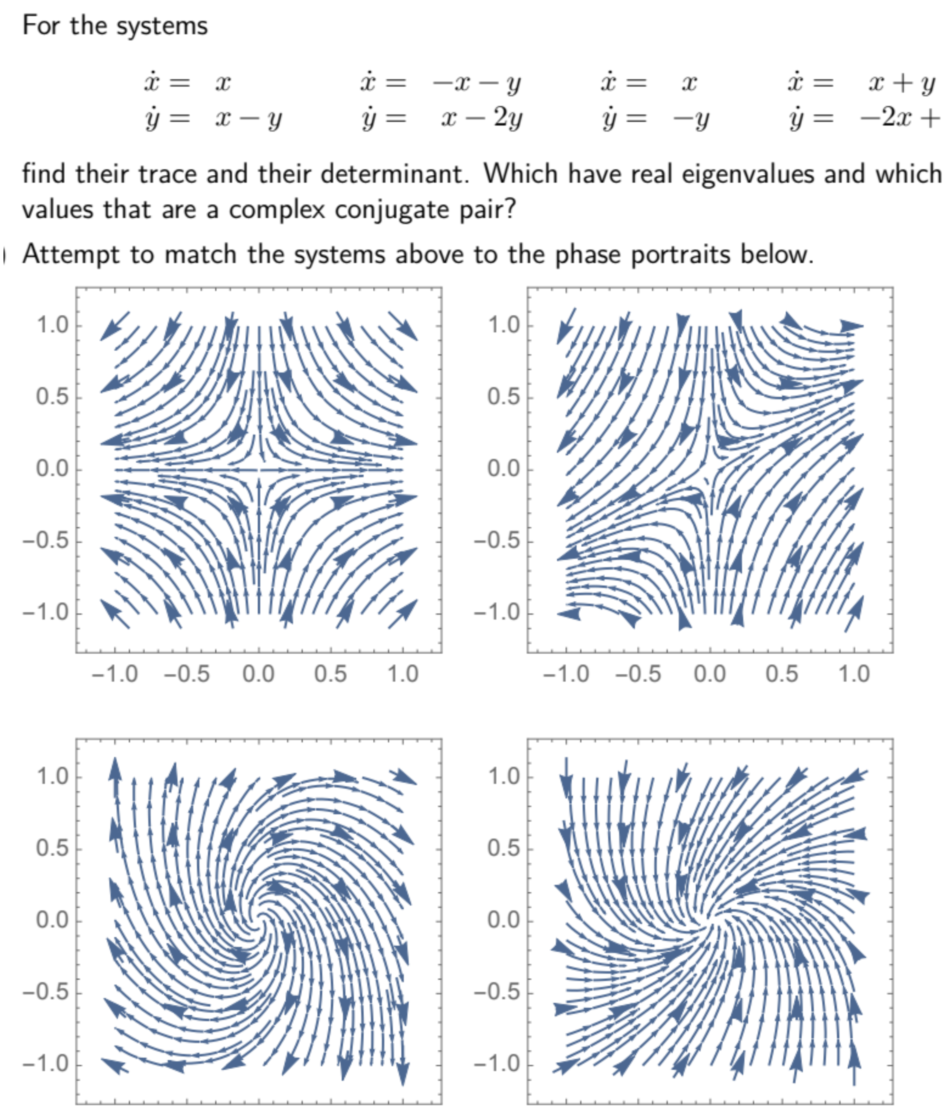
$\lambda_1 = \lambda + i\omega$ and $\lambda_2 = \lambda - i\omega$. When they are a complex conjugate pair, we require the solution to the differential equation to be real, so we require c_1 and c_2 to be a complex conjugate pair as well.

For the systems

$$\begin{array}{cccc} \dot{x} = x & \dot{x} = -x - y & \dot{x} = x & \dot{x} = x + y \\ \dot{y} = x - y & \dot{y} = x - 2y & \dot{y} = -y & \dot{y} = -2x + y \end{array}$$

find their trace and their determinant. Which have real eigenvalues and which have eigenvalues that are a complex conjugate pair?

(g) Attempt to match the systems above to the phase portraits below.


For the systems

$$\begin{array}{llll} \dot{x} = \ x & \dot{x} = \ -x - y & \dot{x} = \ x & \dot{x} = \ x + y \\ \dot{y} = \ x - y & \dot{y} = \ x - 2y & \dot{y} = \ -y & \dot{y} = \ -2x + \end{array}$$

find their trace and their determinant. Which have real eigenvalues and which values that are a complex conjugate pair?

Attempt to match the systems above to the phase portraits below.

Solution. Write

$$\underline{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

(a) The two scalar equations combine into

$$\boxed{\frac{d\underline{x}}{dt} = A\underline{x}}, \qquad \frac{d}{dt}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}.$$

(b) An eigenvalue is, by definition, a value of λ for which $A - \lambda I$ is singular. Therefore the eigenvalues are precisely the roots of the characteristic polynomial

$$p(\lambda) = \det(\lambda I - A) = \lambda^2 - \tau\lambda + \Delta.$$

A monic quadratic with roots λ_1 and λ_2 factors as

$$p(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2),$$

so the equation $p(\lambda) = 0$ is equivalently $(\lambda - \lambda_1)(\lambda - \lambda_2) = 0$.

(c) Expanding the factored form gives

$$(\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2.$$

Comparing this with $\lambda^2 - \tau\lambda + \Delta$ yields

$$\boxed{\lambda_1 + \lambda_2 = \tau}, \qquad \boxed{\lambda_1\lambda_2 = \Delta}.$$

(d) The eigenvalues are

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}.$$

Thus the parabola $\Delta = \tau^2/4$ separates real eigenvalues from complex conjugate eigenvalues. The requested real-eigenvalue regions are

region in the (Δ, τ)-plane	eigenvalue signs
$\Delta < 0$	one positive and one negative
$0 < \Delta \leq \tau^2/4,\ \tau > 0$	two positive
$0 < \Delta \leq \tau^2/4,\ \tau < 0$	two negative

On $\Delta = 0$, one eigenvalue is zero. Above the parabola, $\Delta > \tau^2/4$, the eigenvalues are complex and have real part $\tau/2$. Equality on the parabola gives a repeated real eigenvalue.

(e) Since $A\underline{v} = \lambda_1\underline{v}$,

$$\frac{d}{dt}\left(\underline{v}e^{\lambda_1 t}\right) = \lambda_1\underline{v}e^{\lambda_1 t} = (A\underline{v})e^{\lambda_1 t} = A\left(\underline{v}e^{\lambda_1 t}\right).$$

Hence $\underline{x}(t) = \underline{v}e^{\lambda_1 t}$ solves $\dot{\underline{x}} = A\underline{x}$. If λ_1 is real, every point of the solution is a scalar multiple of the fixed vector $\underline{v}$. Its trajectory therefore lies on the straight line $\operatorname{span}\{\underline{v}\}$. When $\lambda_1 < 0$ it approaches the origin in forward time; when $\lambda_1 > 0$ it moves away from the origin.

(f) For each displayed system, form its coefficient matrix and use $D = \tau^2 - 4\Delta$:

system	A	τ	Δ	$\lambda_{1,2}$
1	$\begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$	0	-1	$1, -1$
2	$\begin{pmatrix} -1 & -1 \\ 1 & -2 \end{pmatrix}$	-3	3	$\dfrac{-3 \pm i\sqrt{3}}{2}$
3	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	0	-1	$1, -1$
4	$\begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix}$	2	3	$1 \pm i\sqrt{2}$

Systems 1 and 3 have real eigenvalues of opposite sign, so both are saddles. System 2 has a complex conjugate pair with negative real part, so it is a stable spiral. System 4 has a complex conjugate pair with positive real part, so it is an unstable spiral.

(g) The matches, reading the portraits from left to right and then top to bottom, are

portrait	system	reason
upper left	3	axis-aligned saddle
upper right	1	tilted saddle; unstable line $y = x/2$
lower left	4	clockwise unstable spiral
lower right	2	counterclockwise stable spiral

For system 1, the stable eigendirection is the y-axis and the unstable eigendirection is $y = x/2$, producing the tilted saddle. For system 3, both eigendirections are coordinate axes. To distinguish the spirals, examine the vector at $(1, 0)$: system 2 gives $(-1, 1)$, which points inward and counterclockwise, while system 4 gives $(1, -2)$, which points outward and clockwise.

□

Answers: 2a: $\underline{x} = \binom{x}{y}$, $\dot{\underline{x}} = \dfrac{d}{dt}\binom{x}{y}$, $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The equation becomes $\dot{\underline{x}} = A\underline{x}$.

2b: $(\lambda - \lambda_1)(\lambda - \lambda_2) = 0$ because the eigenvalues λ_1 and λ_2 are the roots of the characteristic equation.

2c: Expanding, $\lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2 = 0$. We have $\lambda^2 - \tau\lambda + \Delta$ and $\lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2$. These polynomials have the same roots and the same leading coefficient: they are the same polynomial. So $\tau = \lambda_1 + \lambda_2$ and $\Delta = \lambda_1\lambda_2$.

2d: The left side has $\Delta < 0$ so one positive and one negative. The first quadrant has two positive eigenvalues. The fourth quadrant has two negative.

2e: $\underline{x} = \underline{v}e^{\lambda_1 t}$. $\dot{\underline{x}} = \underline{v}\lambda_1 e^{\lambda_1 t}$. $A\underline{x} = A\underline{v}e^{\lambda_1 t}$. $\underline{v}$ is an eigenvector of A so $A\underline{v} = \lambda_1\underline{v}$. The sides match. This is a straight line solution because the solution is a constant multiple of a single vector direction, so we move along that direction (either exponential growth, or exponential decay) as time increases.

2f:

A	τ	Δ and eigenvalues
$\begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$	$1 + (-1) = 0$	$1(-1) - 0(1) = -1 < 0,$ real
$\begin{pmatrix} -1 & -1 \\ 1 & -2 \end{pmatrix}$	$-1 + (-2) = -3$	$2 - (-1) = 3,\ \tau^2 - 4\Delta = -3,$ complex
$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$1 + (-1) = 0$	$1(-1) - 0(0) = -1 < 0,$ real
$\begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix}$	$1 + 1 = 2$	$1(1) - 1(-2) = 3,\ \tau^2 - 4\Delta = -8,$ complex

where $\lambda_{\pm} = \tau/2 \pm \frac{1}{2}\sqrt{\tau^2 - 4\Delta}$.

2g: $\begin{array}{l} \dot{x} = x \\ \dot{y} = -y \end{array}$ has eigenvectors along the axes so matches to the upper left plot.

$\begin{array}{l} \dot{x} = x \\ \dot{y} = x - y \end{array}$ is the other saddle point so matches to the upper right plot.

Hint. For the remaining two with complex conjugate eigenvalues, think about the sign of the real part. Also, how does a (relatively) larger imaginary component influence the shape of the trajectory?